

Calibration results

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Camera-system parameters:

cam0 (/camera_0/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.79131412 0.60602318 1348.64071851 1347.19760585 927.22064613

615.4602873] +- [0.00328698 0.01588842 0.20008261 0.19878619 0.10461723 0.12811622]

reprojection error: [0.000000, -0.000001] +- [0.402630, 0.455358]

cam1 (/camera_1/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.76941645 0.6044969 1331.97707536 1330.42193149 903.03571543

599.75445789] +- [0.00377972 0.0182251 0.260057 0.25804729 0.15256695 0.15281741]

reprojection error: [-0.000000, -0.000001] +- [0.365603, 0.433215]

cam2 (/camera_2/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [3.00062623 0.34547837 1416.05875798 1413.5822091 981.76803234

578.83993784] +- [0.00335285 0.01801893 0.20951746 0.20752544 0.12608249 0.12260964]

reprojection error: [0.000000, -0.000000] +- [0.313368, 0.337868]

cam3 (/camera_3/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.99124545 0.28453608 1413.60774898 1413.05163639 908.51320935

595.8963929] +- [0.00316135 0.01644162 0.18090801 0.17972542 0.09314221 0.11325505]

reprojection error: [-0.000001, -0.000000] +- [0.313663, 0.344606]

baseline T_1_0:

q: [-0.00455854 -0.71786421 -0.00441274 0.69615423] +- [0.00047944 0.00072989 0.00044866]

t: [0.04695634 0.00029887 -0.05007374] +- [0.00014441 0.00006954 0.0001528]

baseline T_2_1:

q: [0.00487115 -0.69754892 -0.00814507 0.7164743] +- [0.0004716 0.00077845 0.0005036]

t: [0.04820178 0.0005072 -0.04864423] +- [0.00019044 0.00007533 0.00016947]

baseline T_3_2:

q: [0.00325727 -0.70372973 -0.00034306 0.71046023] +- [0.0004077 0.00061418 0.00043423]

t: [0.04824171 0.0003938 -0.0484075] +- [0.00012058 0.00004991 0.00011335]

Target configuration

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Type: aprilgrid

Tags:

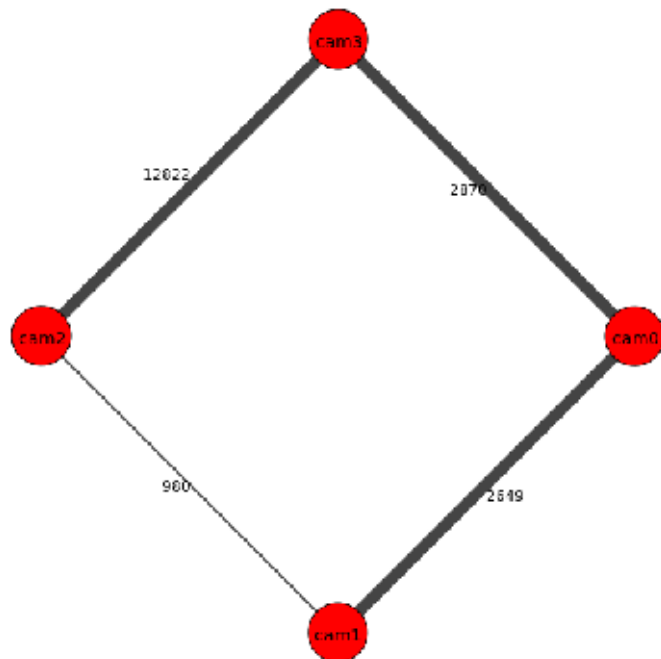
Rows: 6

Cols: 6

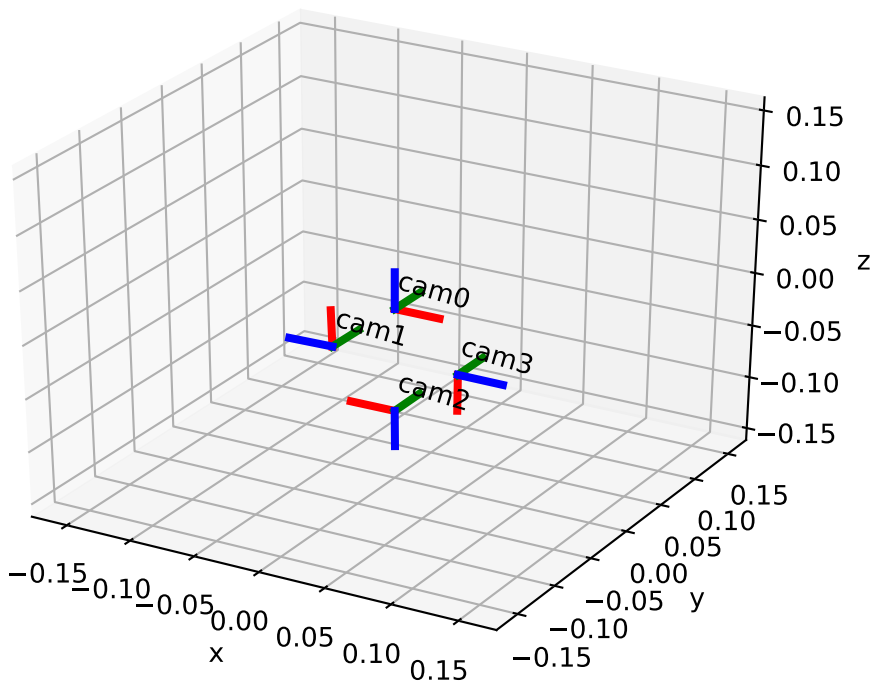
Size: 0.021 [m]

Spacing 0.0059997 [m]

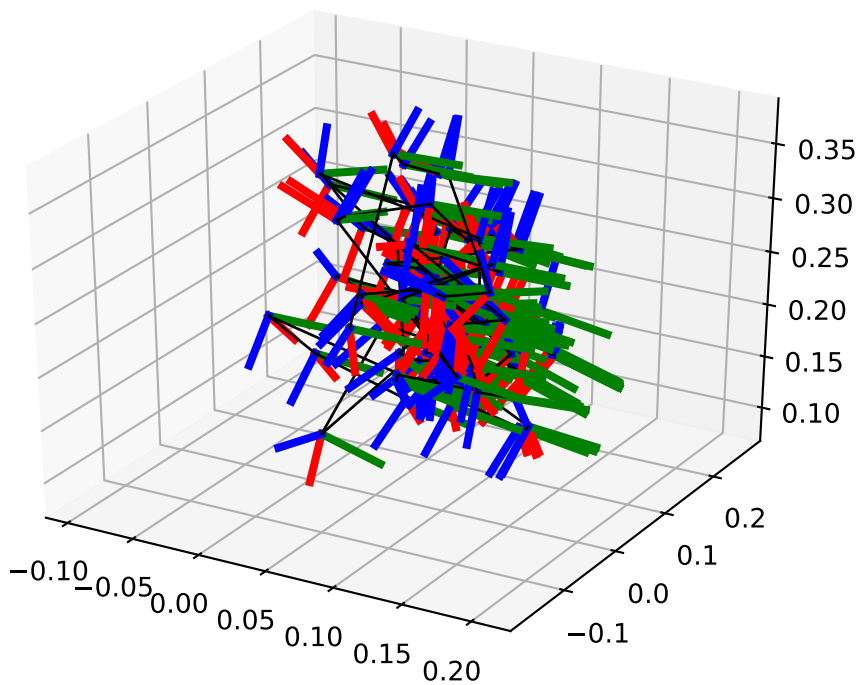
Inter-camera observations graph (edge weight=#mutual obs.)



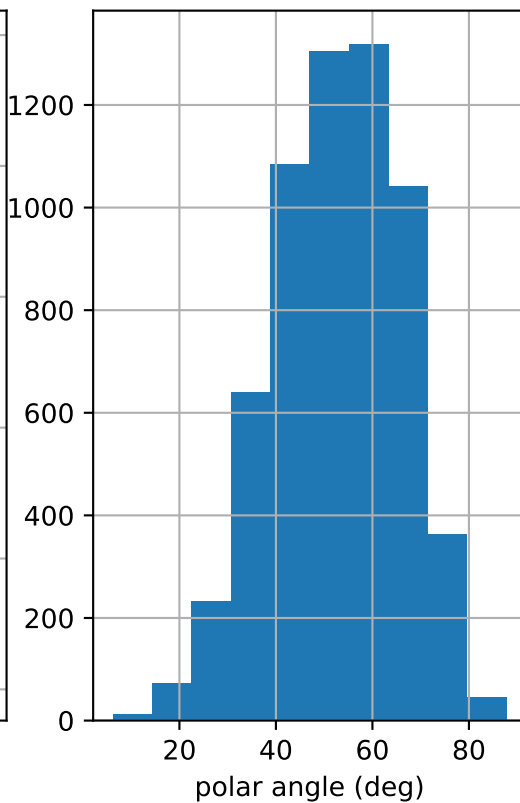
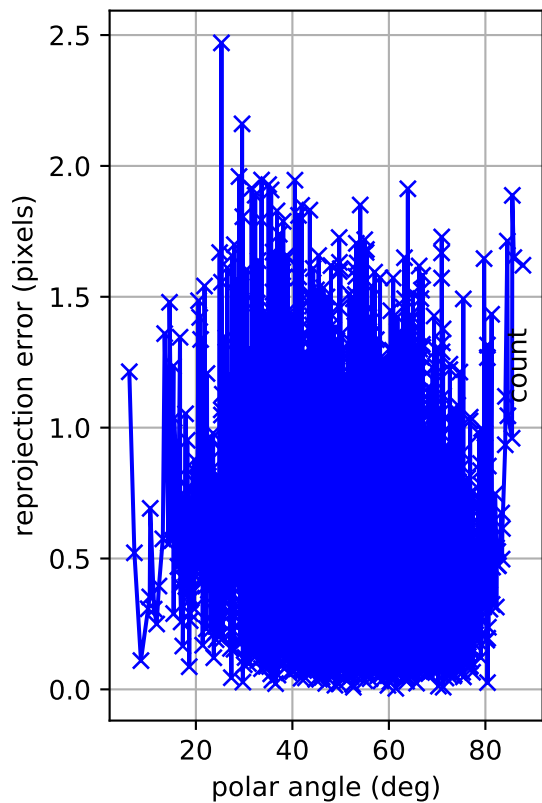
camera system



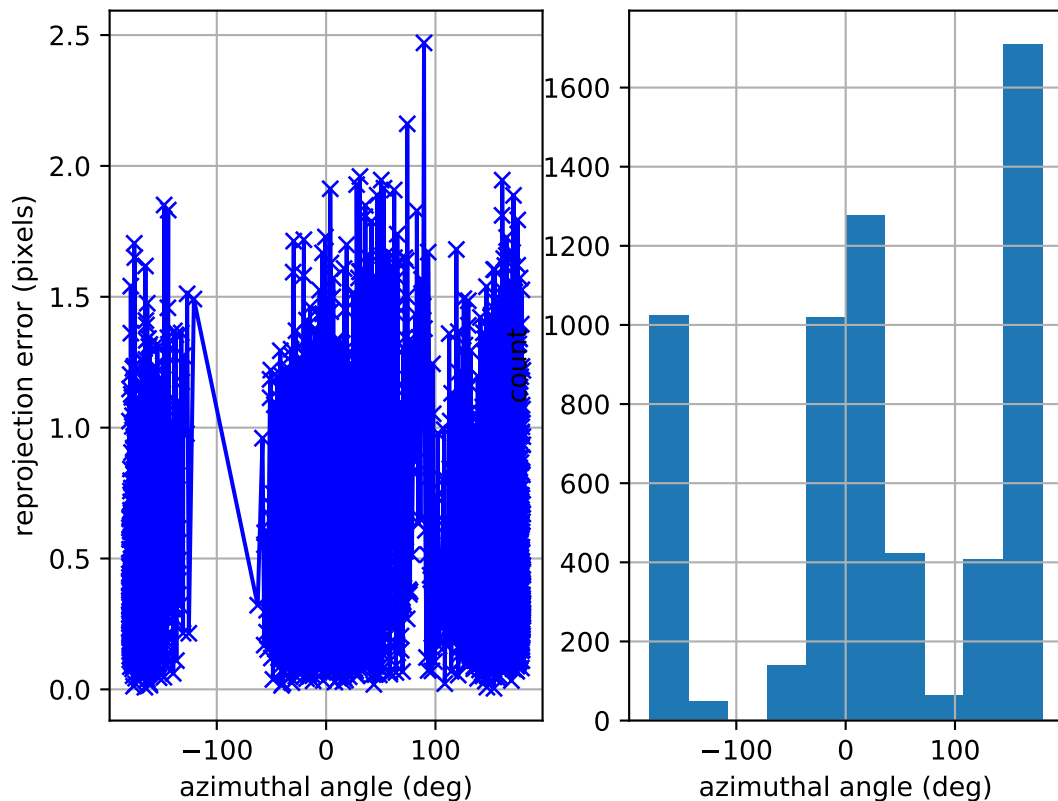
cam0: estimated poses



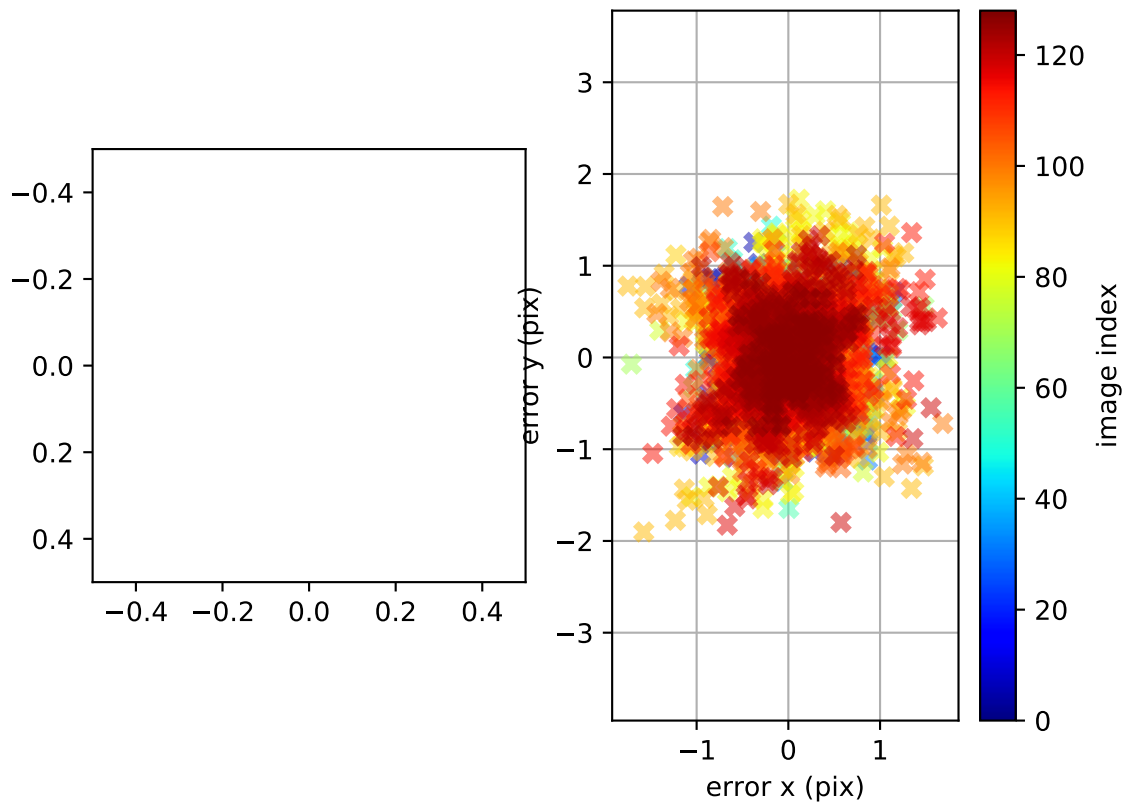
cam0: polar error



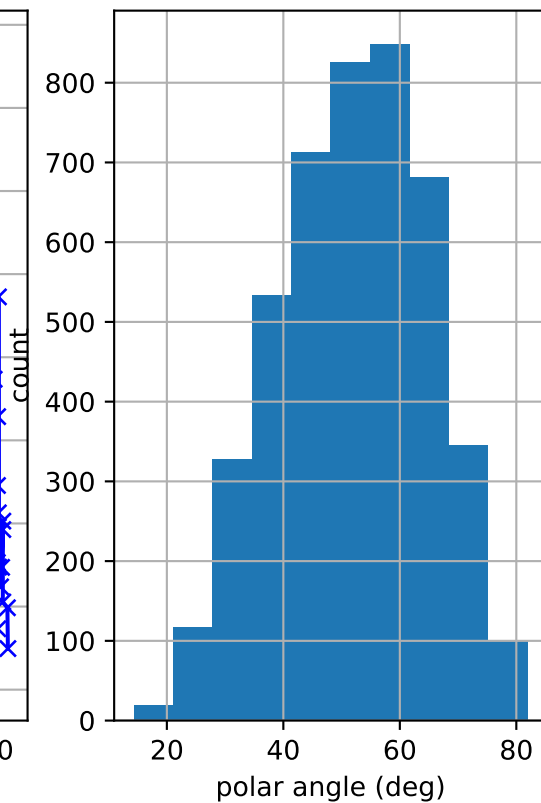
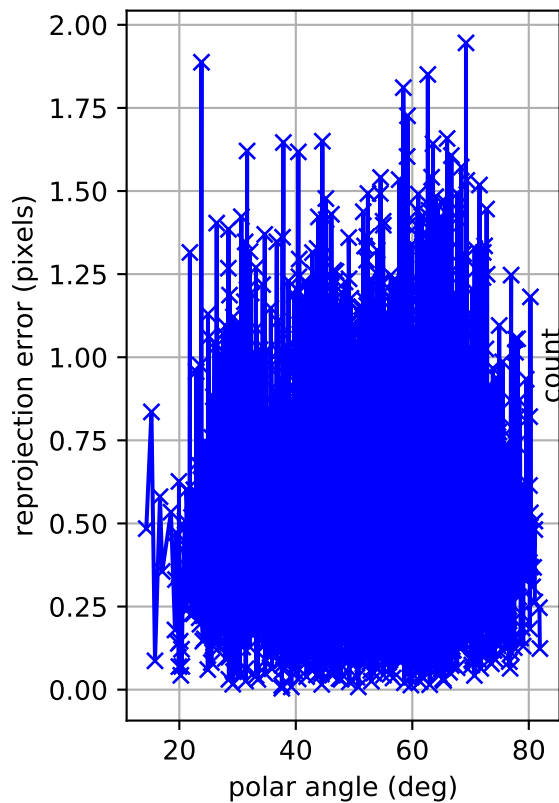
cam0: azimuthal error



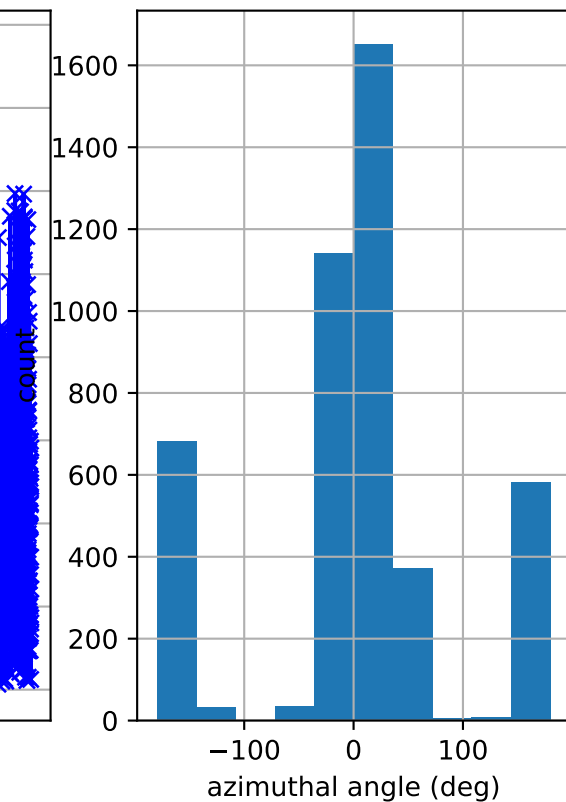
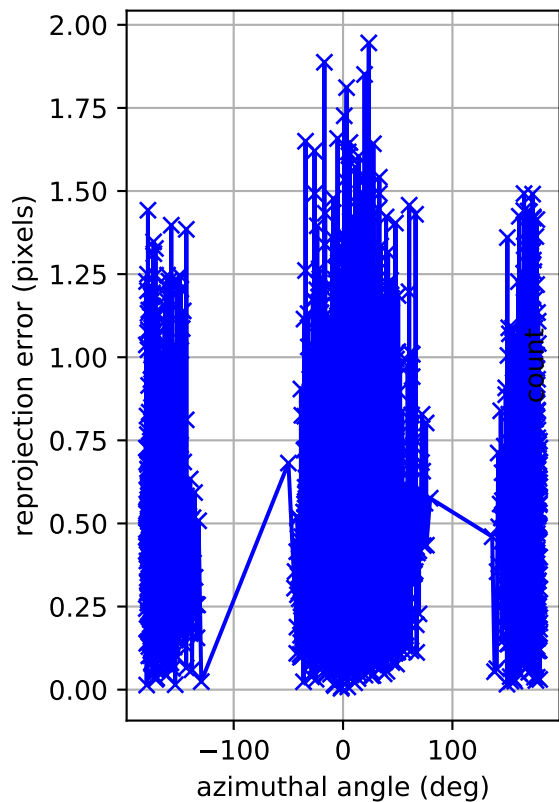
cam0: reprojection errors



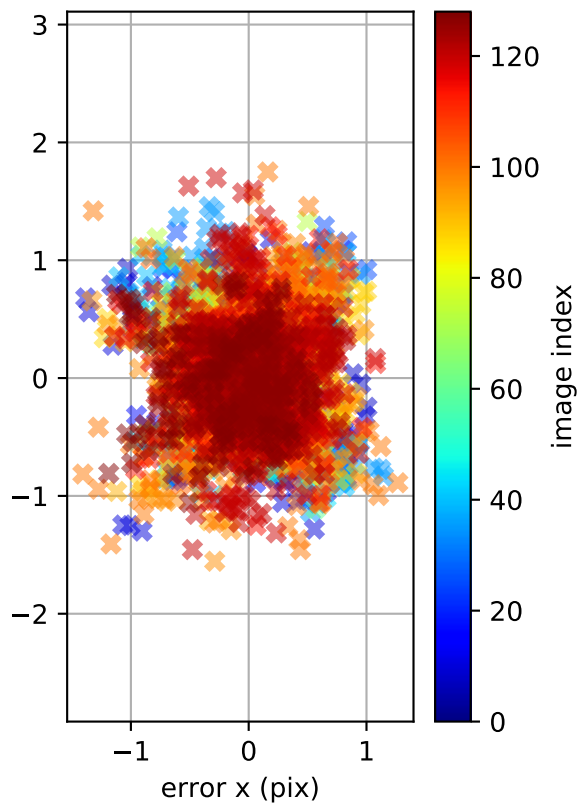
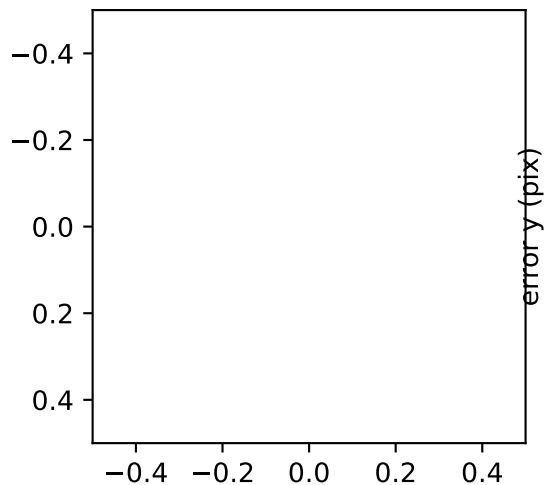
cam1: polar error



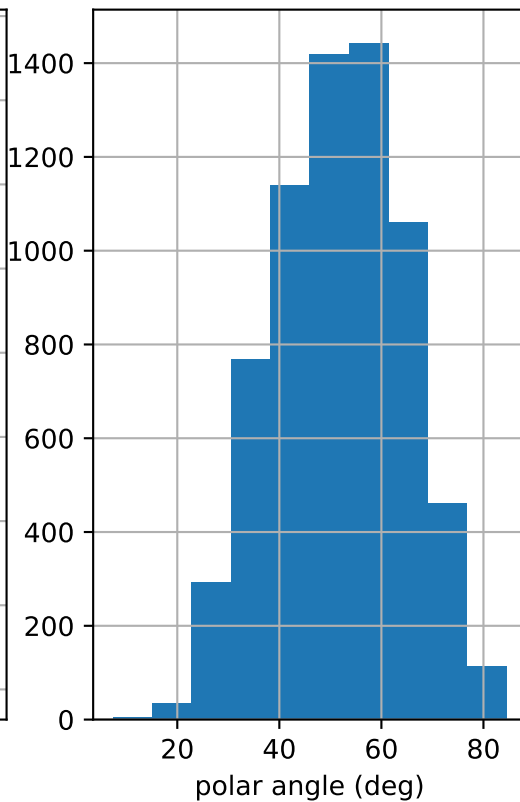
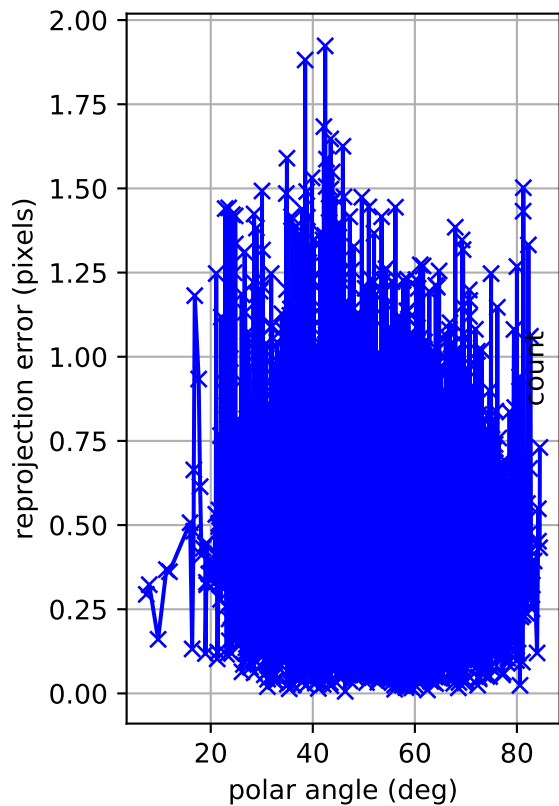
cam1: azimuthal error



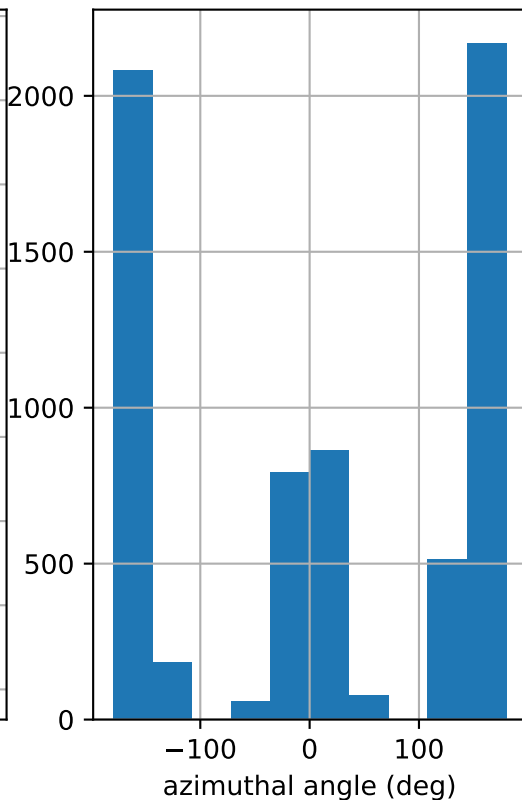
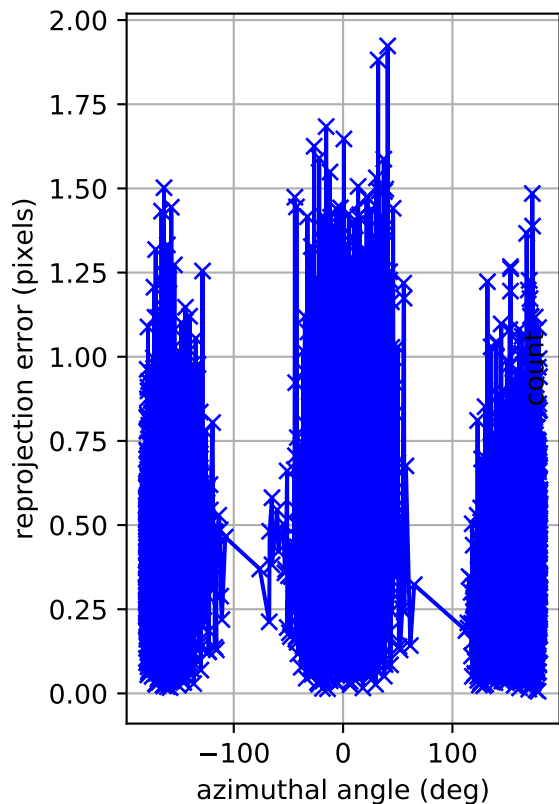
cam1: reprojection errors



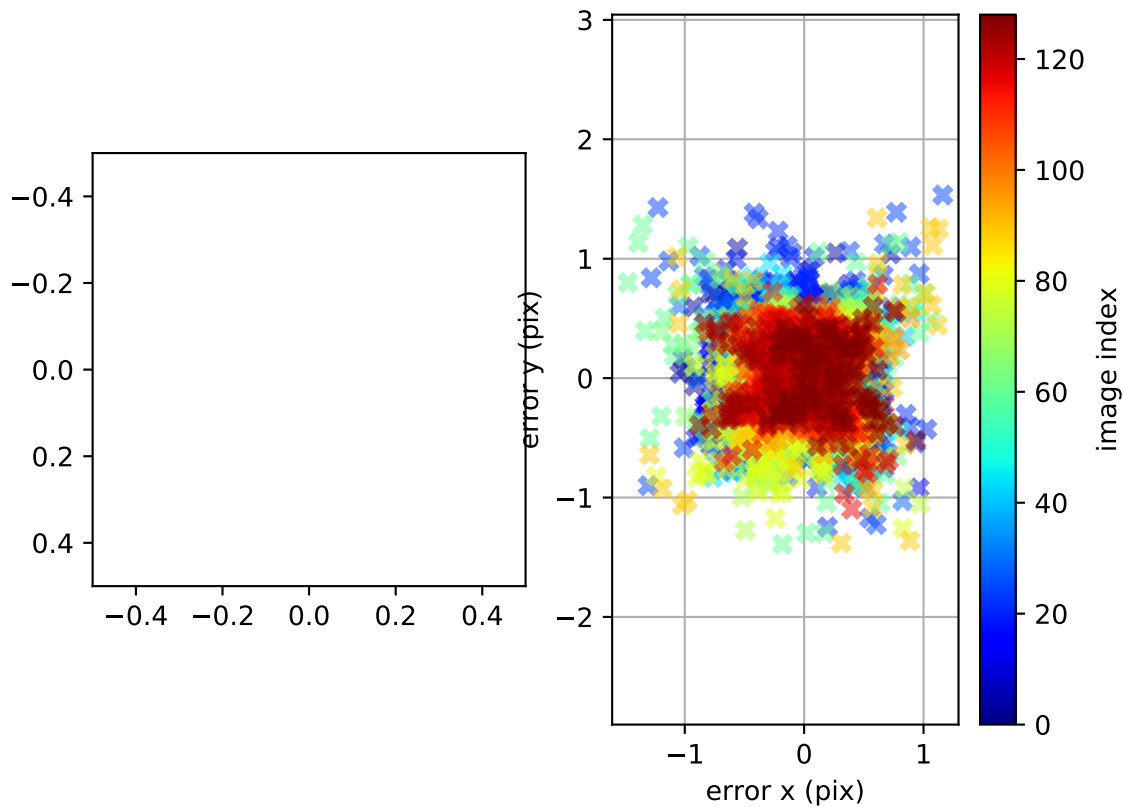
cam2: polar error



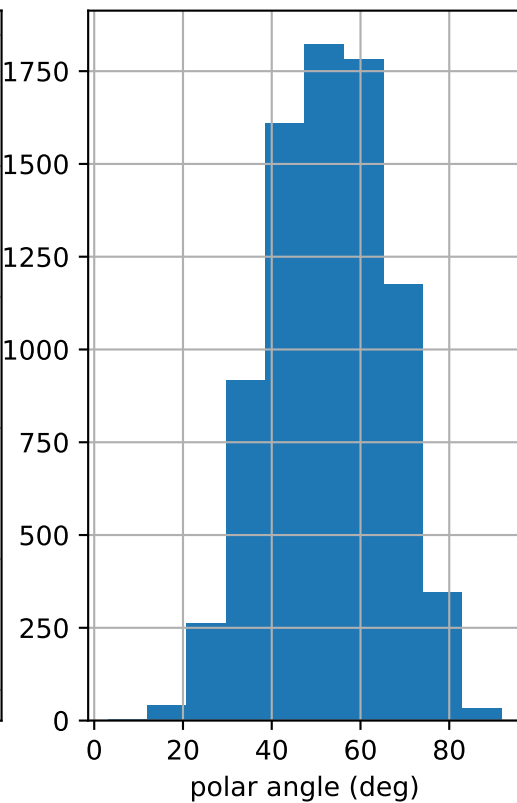
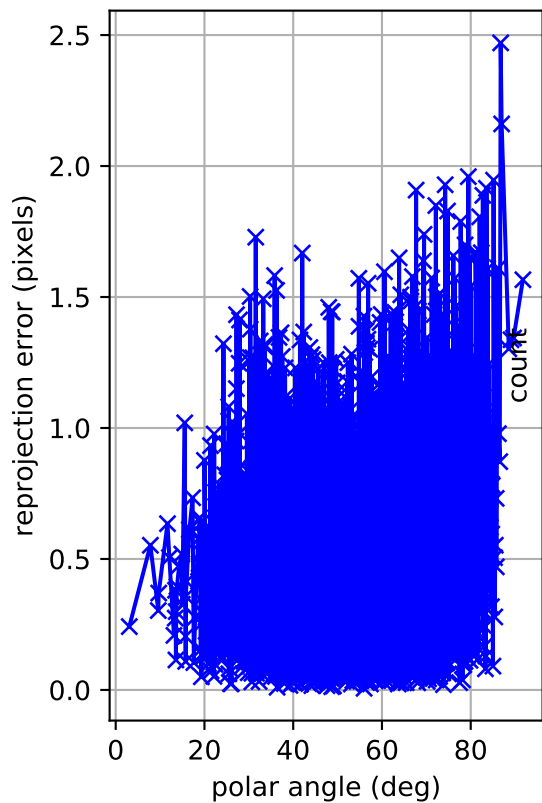
cam2: azimuthal error



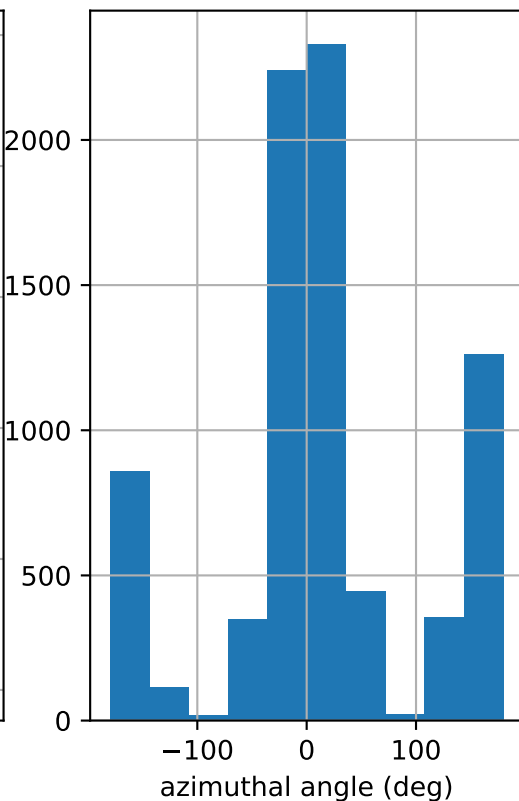
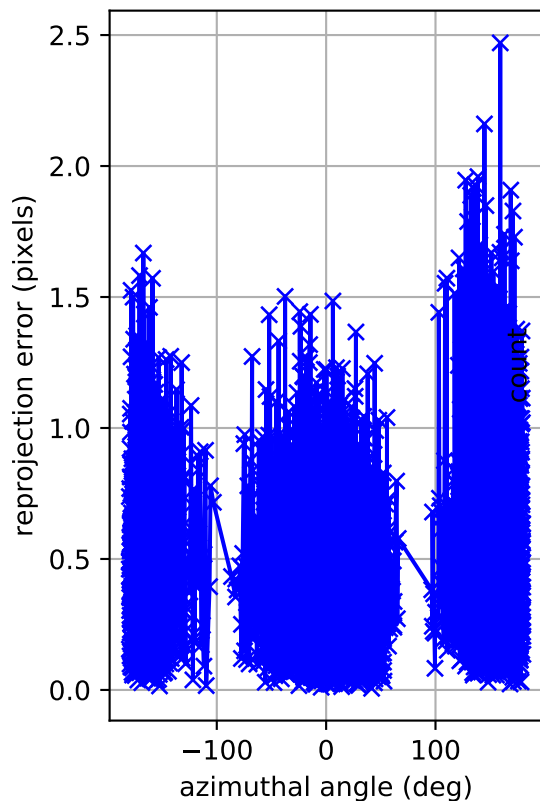
cam2: reprojection errors



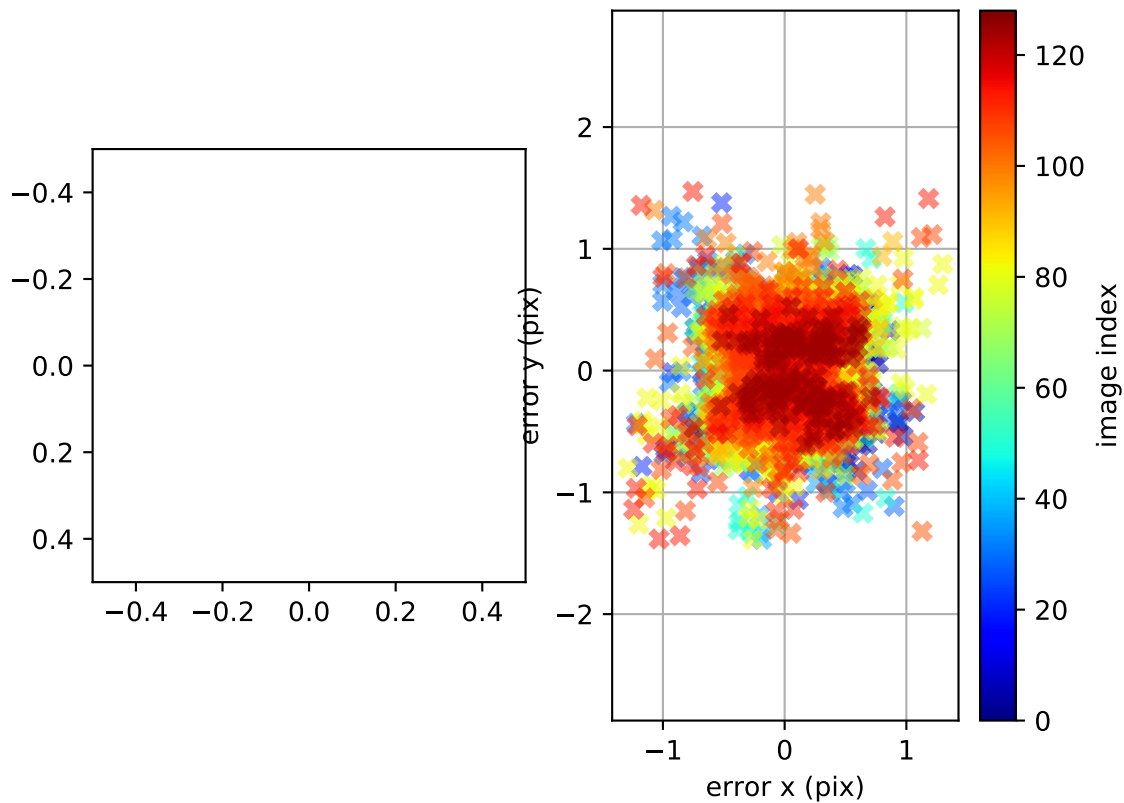
cam3: polar error



cam3: azimuthal error

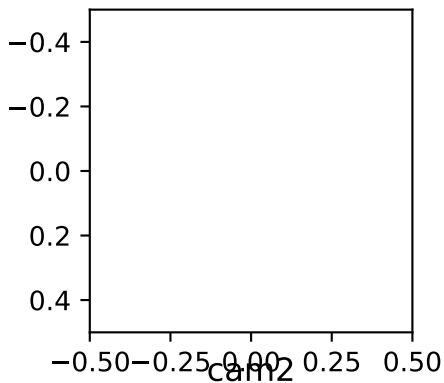


cam3: reprojection errors

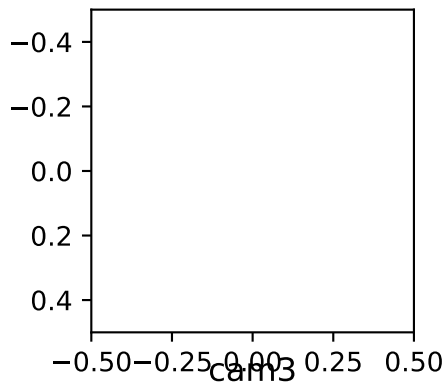


Location of removed outlier corners

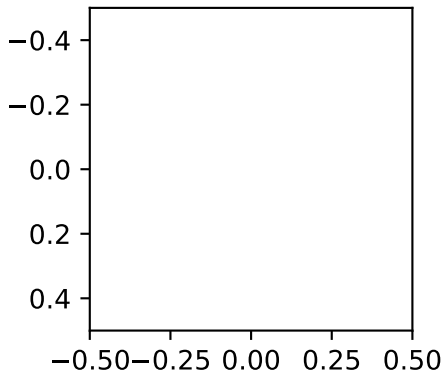
cam0



cam1



cam2



cam3

